Response to Reviewers

Association between fridge opening frequency and probability of finding something new inside

We thank both reviewers, and the editor, for their careful reading of the manuscript. Below we address each comment in turn, with the corresponding change cited by page number in the manuscript.

Reviewer 1

Reviewer 1 — comment 1

"Fridge-checking" needs an operational definition, not just an intuitive label — otherwise readers cannot tell it apart from ordinary refrigerator use.

We have added an explicit definition, following the Delphi consensus terminology of Ware and Crumb.

p. 2 — We define fridge-checking , following the terminology introduced by [1], as any refrigerator-door-opening event that is not immediately followed by either the removal or the addition of a food item.

Reviewer 1 — comment 2

A single logistic regression with one observation per household seems to discard information — weren't opening events logged daily over 30 days per household?

Correct, and thank you for catching this. We now use generalized estimating equations to account for the repeated, within-household structure of the data, rather than collapsing to one observation per household.

p. 3 — ~~The association between opening frequency and discovery probability was assessed using ordinary logistic regression, with one observation per household.~~ The association between opening frequency and discovery probability was assessed using generalized estimating equations with an exchangeable working correlation structure [2], to account for the repeated, within-household structure of the daily opening logs.

Reviewer 1 — comment 3

Please also update the model equation itself to match the revised analysis — a linear probability model for a probability outcome is not appropriate here.

Done; the model is now written on the logit scale, with an added household-level random effect, numbered (1) in the manuscript:

p. 3 — The primary model

$$\cancel{P(\text{discovery}) = \beta_0 + \beta_1 \cdot \text{openings}}$$

$$\text{logit}(P(\text{discovery})) = \beta_0 + \beta_1 \cdot \text{openings} + u_{\text{household}} \quad (1)$$

~~is a naive linear probability model.~~ , where $u_{\text{household}}$ is a household-level random effect absorbing unmeasured, time-invariant fridge-checking tendencies.

Reviewer 1 — comment 4

How were households actually recruited? "Flyers" alone is a bit vague for a methods section.

We have kept this wording, which we believe is already sufficiently specific once the recruitment venues are listed.

p. 2 — Households were recruited via flyers posted in laundromats, grocery stores, and one particularly well-trafficked gym.

Reviewer 1 — comment 5

Table 1's household-size figure for the high tertile looks inconsistent with the number reported in the abstract — please reconcile.

Thank you for catching this; the table has been corrected.

p. 2 —

	Low tertile	Medium tertile	High tertile
Household size, mean	1.8	2.4	~~2.6~~ 3.1
Owns a pet, %	12	19	31
Fridge magnets, median	3	5	9
Snacks after 10pm, %	21	38	64

Table 1: Household characteristics by tertile of daily fridge-opening frequency.

Reviewer 1 — comment 6

The calibration equation appears to duplicate the main model equation once the model is refit on the logit scale — is it still needed?

Good catch, it is not: once refit with a logit link, the calibration equation collapses to the same closed form as the model equation above, so we have removed it as redundant.

Removed: the closed-form calibration equation, which became algebraically identical to the model equation above once it was refit with a logit link.

Reviewer 1 — comment 7

An analysis of robustness to unmeasured confounding — hunger, in particular, given the subject matter — is missing.

We agree this was a gap and have added a sensitivity analysis for unmeasured hunger confounding, cited in the manuscript. (modified on p. 6)

Reviewer 1 — comment 8

The conclusion's "intermittent-reinforcement device" framing is memorable but should probably be tied more explicitly to the behavioral literature — or else softened.

We have reviewed this passage and, on balance, prefer to keep the wording as originally written; we believe the framing is self-explanatory in context.

p. 7 — Taken together, these findings suggest that the refrigerator door itself may function as an intermittent-reinforcement device, rewarding repeated checking with an occasional, genuinely novel discovery.

Reviewer 1 — comment 9

Please also report sensitivity to the door-handle contamination threshold used to define a "genuine" opening event.

This analysis was performed but, on reflection, is redundant with the hunger-confounding sensitivity analysis and has been removed for concision.

Removed: the sensitivity analysis by handle-contamination threshold.

Reviewer 1 — comment 10

One comment, filed here for convenience even though it concerns two separate locations in the manuscript: the operational definition of fridge-checking, introduced early on, should be explicitly reaffirmed in the conclusion, so that a reader who skips straight to the conclusion isn't left guessing what was actually measured.

Addressed in both locations: (modified on p. 2 and p. 7).

Shown as two separate excerpts, one per location:

p. 2 — This comment from the first reviewer required a correction in two places in the manuscript — this sentence, in the introduction, and a matching one in the conclusion — applies equally to both instances.

p. 7 — As already noted in the introduction, this conclusion applies equally to the paired instance of the same reviewer comment discussed there.

Reviewer 2

Reviewer 2 — comment 1

The "Statistical analysis" heading should indicate that the analysis plan changed relative to what was originally registered.

The heading has been updated accordingly.

p. 2 — Statistical analysis (updated)

Reviewer 2 — comment 2

A subgroup analysis (pet ownership, household size, weekday versus weekend) would help readers judge how broadly the main finding generalizes.

A subgroup figure has been added, reporting the requested comparisons. (modified on p. 4)

Reviewer 2 — comment 3

The sock-drawer control group is an unusual choice and, frankly, distracts from the paper's main contribution — consider removing it given the space constraints.

We agree and have removed this sub-analysis in its entirety, including its figure and table.

Removed: the sock-drawer control group sub-analysis, including its figure and table.

Reviewer 2 — comment 4

The discussion's causal language ("causes an increase") is stronger than the observational design supports.

We have softened this to an associational claim and noted the absence of a formal blinding procedure for refrigerator contents, in the same spirit as our response to reviewer 1, comment 2, p. 1 on accounting properly for the repeated-measures structure of the data.

p. 6 — ~~These results suggest that fridge-checking behavior causes an increase in perceived discovery probability.~~ These results are consistent with an association between fridge-checking behavior and perceived discovery probability, though the observational design and the absence of a formal blinding procedure for refrigerator contents limit causal interpretation.

Reviewer 2 — comment 5

Theorem 1 is charming, but if it is to remain in a revised manuscript it should at least be given a name, so it can be cited by future work.

We have named it the Second Law of Fridge Thermodynamics. (modified on p. 4)

Reviewer 2 — comment 6

Table 2 (fridge discovery categories) should include an "unidentified frozen object" category — in our experience this is the single most common terminal state of a forgotten leftover.

Added, at 3% of day-30 discoveries.

p. 5 —

Category	Day 1	Day 30
Forgotten leftovers	61%	58%
Mystery Tupperware	24%	22%
Something green	9%	7%
Actually edible food	6%	13%
Unidentified frozen object	–	3%

Table 2: Fridge discovery categories, day 1 versus day 30 of observation.

Editor

Editor — comment 1

Please ensure that every figure and table added or modified in response to the reviewers is cited with a page number in this letter, per journal policy.

Done throughout; see in particular the subgroup figure, referenced at Figure 2, p. 4, and the discovery-categories table, referenced at Table 2, p. 5.

Author notes

The changes above address every reviewer and editor comment. Two further, smaller changes were made on our own initiative during revision, noted here for transparency even though neither answers a specific reviewer comment.

Tupper Ware

Tupper Ware — change 1

While implementing the change requested in our response to reviewer 1, comment 2, p. 1, I noticed the household-level random effect variance was worth reporting explicitly, so I added it.

p. 3 — While implementing the change above, we also noted that the estimated household-level random effect variance, $\hat{\sigma}_u^2 = 0.42$, is worth reporting explicitly: it indicates non-trivial between-household heterogeneity in fridge-checking tendencies beyond what opening frequency alone explains.

Ivana Snackwell

Ivana Snackwell — change 1

Re-read the fridge-checking definition after the change above and confirm it still matches our coding procedure.

(see p. 2)

For the reviewers only

The table below summarizes, for the reviewers' convenience only, how many comments from each reviewer led to an actual textual or figure/ table change in the manuscript, as opposed to a clarification with no change — consistent with the general observation that peer review of a self-monitoring behavior tends to attract more critical attention than the behavior itself D. Ajar [3].

Reviewer	Comments	Led to a change
Reviewer 1	10	8
Reviewer 2	6	6

Table R3: Summary of comments and resulting changes, for the reviewers' convenience.

As shown in Table R3, the large majority of comments from both reviewers led directly to a manuscript change, consistent with our own prior finding that fridge-checking behavior is, if nothing else, a reliable trigger for further action I. Snackwell [4].

References cited in this response

[1] T. Ware and N. Crumb, "Defining Fridge-Checking Behavior: A Delphi Consensus Study," *Annals of Domestic Science*, 2019.

[2] G. E. Estimator, "Generalized Estimating Equations for Clustered Snacking Data," *Journal of Applied Statistics*, 2017.

[3] D. Ajar, "The Observer Effect in Refrigerator Peeking," *Journal of Irreproducible Snacking*, 2015.

[4] I. Snackwell, "The Fridge-Checking Phenomenon: A Preliminary Report," *Journal of Kitchen Epidemiology*, 2020.